ABSTRACT OF THE DISCLOSURE

An apparatus and system for a mass storage device that replaces traditional serial data interfaces with a massively parallel optical interface. This Optically-Attached Storage (OAS) device integrates a non-volatile storage medium with a two-dimensional array of photon emitters and sensors for data input/output. By communicating via frames of light rather than serial electrical signals, the OAS device bypasses the I/O bottleneck of standards like SATA, SAS, or NVMe, allowing for data transfer rates limited only by the physical read/write speed of the underlying storage medium. The invention provides a foundational component for next-generation data centers and physical data logistics systems, enabling the ultra-rapid ingest and offload of petabyte-scale datasets.